

Lulu Zhang

Doctor of Philosophy (PhD) in Astrophysics

✉ l.l.zhangastro@gmail.com

📍 UT San Antonio, TX

🏠 luluzhang.myportfolio.com

🆔 ORCID: 0000-0003-4937-9077

Research Interests and Goals

My research focuses on spatially resolved studies of infrared emission features from PAHs, H₂, and ionized gas and their connection to physical conditions across diverse galactic environments, with an emphasis on multi-wavelength diagnostics of star formation and AGN activity and their associated feedback effects, including shocks and outflows. By examining PAH features and other diagnostics, I aim to deepen our understanding of the physical processes behind AGN feedback and how they shape galaxy evolution.

Education and Professional Appointments

Research Assistant Professor <i>UT San Antonio, San Antonio, USA</i>	2026 Sep –
Postdoctoral Researcher <i>UT San Antonio, San Antonio, USA</i> Faculty Host: Prof. Chris Packham	2023 – 2026
Ph.D., Astrophysics <i>Peking University, Beijing, China</i> Dissertation: <i>The Spatially Resolved Mid-infrared Spectral Diagnostics in Nearby Galaxies</i> Advisor: Prof. Luis C. Ho	2017 – 2023
B.S., Applied Physics <i>Shandong University, Weihai, China</i>	2013 – 2017

Publication Summary

See the full publication list on [ADS](#), or ORCID [0000-0003-4937-9077](#)
33 papers in total (10 first-author) with over 500 citations (ADS; as of Sep 2026)
h-index: 13 (as of Sep 2026)

Approved Proposals

ISSI International Team Call 2026 (26-681; invited member) <i>PAHs Across Cosmic Environments: JWST and Galaxy Evolution</i>	2026 <i>Duration: 2026-2028</i>
JWST Cycle 4 (GO-07429; P.I.) <i>Unraveling Effects of Different Modes of AGN Feedback via PAH Features</i>	2025 <i>Allocation: 11.7 hours</i>
JWST Cycle 4 (GO-07802; Co-I.) <i>Dust in shocks: The missing link in AGN feedback</i>	2025 <i>Allocation: 14.2 hours</i>
JWST Cycle 4 (GO-07195; Co-I.) <i>Deciphering the torus and extended dust properties of local active galactic nuclei</i>	2025 <i>Allocation: 32.8 hours</i>
JWST Cycle 3 (GO-04972; P.I.) <i>Hunting the Kinetic Mode Feedback of AGNs via PAH Features</i>	2024 <i>Allocation: 12.4 hours</i>
JWST Cycle 3 (GO-05017; Co-I.) <i>AGN and their outflows: probing fragmentation and survival of PAH molecules</i>	2024 <i>Allocation: 18.7 hours</i>
JWST Cycle 2 (DD-06659; Co-I.) <i>Dust to Dust: Probing the Survival of Cold Dust, New Hot Dust Formation, and Mass-Loss History of the Nearest Type Ibn Supernova</i>	2024 <i>Allocation: 6.3 hours</i>

Research Funding

Space Telescope Science Institute (JWST-GO-07429; P.I.) <i>Unraveling Effects of Different Modes of AGN Feedback via PAH Features</i>	2025 – <i>Amount: \$ 120.2 k</i>
Space Telescope Science Institute (JWST-GO-04972; P.I.) <i>Hunting the Kinetic Mode Feedback of AGNs via PAH Features</i>	2024 – <i>Amount: \$ 129.7 k</i>

Selected Awards

- 2024– Postdoctoral Research Fellow of the Year, UTSA
2017–23 Award for Outstanding Doctoral Graduates of PKU
Award for Outstanding Doctoral Dissertation of PKU (Advisor: Prof. Luis C. Ho)
2013–17 Award for Outstanding Undergraduates of SDU, Weihai
Award for Outstanding Undergraduates of Shandong Province
Award for Outstanding Bachelor's Thesis of SDU (Advisor: Prof. Lulu Fan)

Service and Outreach

- 2026 Invited Judge for “UTSA Postdoctoral Research Fellow of the Year Award”
2026–28 Invited International Space Science Institute International Team Member
2025 Invited NASA/ROSES Review Panelist
2024 Invited Judge for “UTSA Final 3MT Research Competition”
2024–25 HWO AGN Work Group Steering Committee Member (sub-group leader)
2023– TMT International Science Development Team Member
2023– GATOS Collaboration Associate then Full Member
2022– Referee for AJ, ApJ, ApJL, A&A, and MNRAS
2019–22 Cashier for Graduate Fund (Donated by Graduates) of KIAA
2019–20 Admin for Astro-wiki Page of a Biweekly ISM&Galaxy Group Meeting at KIAA
2019 LOC for International Conference – “Cosmic Evolution of Quasars” at KIAA
2018–19 Organizer for Weekly “Happy Hour” of KIAA
2018/19 Secretary for Graduate Student Interview of KIAA

Teaching and Mentoring

Part-Time Lecturer (Undergraduate Course) – AST1013: Introduction to Astronomy

UT San Antonio

Spring 2026

- 2.5 hours lectures on Jovian Planet Systems and 2.5 hours lectures on Lives of Stars

Teaching Assistant (Graduate Course) – Astrophysics of Diffuse Interstellar Medium

Peking University

Spring 2020

★Students Advised

John Schneider, graduate student at UT San Antonio, as Ph.D. Committee Member

2024–

William Olson, undergraduate student at UT San Antonio

2024–

Seminars, Conferences, & Workshops

- 2026 July “Young Astronomers Learning LSST (YALL) workshop”, College Station, USA
2026 May “Torus Workshop 2026: Multi-wavelength Approach to The AGN Structures”, Kyoto, Japan (oral talk)
2026 Apr “COS Symposium, UTSA”, San Antonio, USA (oral talk)
2026 Apr “Physics and Astronomy Seminar, Mizzou”, Online (virtual talk; invited)
2026 Jan “Physics and Astronomy Seminar, UTSA”, San Antonio, USA (oral talk)
2026 Jan “Inaugural Session of NASA AGN Science Interest Group Spotlight Series”, Online (virtual talk; invited)
2025 Nov “The Fifth Pan Dust Conference”, Tucson, USA (poster talk)
2025 Oct “Dust and Gas in the HST and JWST Era”, Toruń (hybrid), Poland (virtual talk; invited)
2025 Oct “Infrared Spectroscopy from Space: New Frontiers from Exoplanets to the Early Universe”, Pasadena, USA (oral talk)
2025 June “AAS 246th Annual Meeting”, Anchorage, USA (oral talk & judge of poster section)
2025 May “GATOS Meeting - Tenerife 2025”, Tenerife, Spain (oral talk)
2024 Nov “HWO AGN Working Group Presentation”, Online (virtual talk; invited)
2024 Jan “Extra Galaxy Evolution Seminar, Oxford”, Oxford, UK (oral talk)
2024 Jan “GATOS Workshop - Madrid 2024”, Madrid, Spain (oral talk)

- 2024 Jan “GATOS Meeting - Madrid 2024”, Madrid, Spain (oral talk)
- 2023 Nov “Physics and Astronomy Seminar, UTSA”, San Antonio, USA (oral talk)
- 2023 Sep “The First Year of JWST Science Conference”, Baltimore, USA (poster talk)
- 2023 May “The 25th Guoshoujing Workshop and the Galaxy & Cosmology Frontier Workshop”, Huangshan, China (oral talk, session chair)
- 2021 Dec “Chinese Astronomical Society Annual Meeting”, Nanchong, China (oral talk)
- 2021 Nov “KIAA Forum on Gas in Galaxies: for Early Career Scientists”, Online (oral talk)
- 2021 Oct “East Asia AGN Workshop 2021”, Chongqing, China (poster)
- 2021 June “The 11th C-MaNGA Workshop and the CSST-IFS Science Meeting”, Zhuhai (postponed), China (oral talk, session chair)
- 2021 May “The 23th Guoshoujing Workshop and the Galaxy & Cosmology Frontier Workshop”, Hangzhou, China (poster)
- 2020 Oct “Supermassive Black Hole Workshop”, Lijiang, China (oral talk)
- 2019 Sep “KIAA Forum on Gas in Galaxies: Probing the Activities and Power Engines from Local to Distant Universe”, Beijing, China (poster)
- 2019 Sep “Chinese Astronomical Society Annual Meeting”, Delingha, China (oral talk)
- 2019 Jan “East Asia AGN Workshop 2019”, Taipei, China (poster)
- 2018 July “Thirty Meter Telescope-MICHI and China: a Workshop on Thermal-IR Astronomy”, Beijing, China (oral talk)
- 2018 July “The 20th Guoshoujing Workshop and the Galaxy & Cosmology Frontier Workshop 2018”, Sanmenxia, China (poster)

Publication List

★ As first-author:

- [10] **Zhang, L.**, Packham, C., Hicks, K. S. E., et al., “GATOS: Distinct Feedback Modes in AGN Central Regions Revealed by Spatially Resolved JWST Spectroscopy”, 2026, ApJL, 1007, L52
- [9] **Zhang, L.**, Packham, C., Hicks, K. S. E., et al., “Evidence of Feedback Effects in Low-luminosity Active Galactic Nuclei Revealed by JWST Spectroscopy”, 2026, ApJL, 998, L32
- [8] **Zhang, L.**, Davies, I. R., Packham, C., et al., “Theoretical Diagnostics for the Physical Conditions in Active Galactic Nuclei under the View of JWST”, 2025, ApJS, 280, 65
- [7] **Zhang, L.**, García-Bernete, I., Packham, C., et al., “Polycyclic Aromatic Hydrocarbon Emission in the Central Regions of Three Seyferts and the Implication for Underlying Feedback Mechanisms”, 2024, ApJL, 975, L2
- [6] **Zhang, L.**, Packham, C., Hicks, K. S. E., et al., “The Galaxy Activity, Torus, and Outflow Survey (GATOS). IV. Exploring Ionized Gas Outflows in Central Kiloparsec Regions of GATOS Seyferts”, 2024, ApJ, 974, 195
- [5] **Zhang, L.**, & Ho, L. C., “The Interaction between AGN and Starburst Activity in the Circumnuclear Region of NGC 7469 as Viewed with JWST”, 2023, ApJL, 953, L9
- [4] **Zhang, L.**, & Ho, L. C., “Estimating Molecular Gas Content in Galaxies from Polycyclic Aromatic Hydrocarbon Emission”, 2023, ApJ, 943, 1
- [3] **Zhang, L.**, & Ho, L. C., “The Contribution of Evolved Stars to Polycyclic Aromatic Hydrocarbon Heating and Implications for Estimating Star Formation Rates”, 2023, ApJ, 943, 60
- [2] **Zhang, L.**, Ho, L. C., & Li, A., “Evidence that Shocks Destroy Small PAH Molecules in Low-luminosity Active Galactic Nuclei”, 2022, ApJ, 939, 22
- [1] **Zhang, L.**, Ho, L. C., & Xie, Y., “A Method to Extract Spatially Resolved Polycyclic Aromatic Hydrocarbon Emission from Spitzer Spectra: Application to M51”, 2021, AJ, 161, 29

† As significant contributing-author:

- [3] Schneider, J., Packham, C., **Zhang, L.**, et al., “Torus Models of NGC 4388: an Elevated MIR/submm Emission Ratio”, 2026, ApJ, submitted
- [2] Delaney, E. D., Hicks, K. S. E., **Zhang, L.**, et al., “GATOS XIII. Outflow Driven Excitation in Seyferts”, 2026, ApJ, 1002, 20
- [1] Delaney, E. D., Hicks, K. S. E., **Zhang, L.**, et al., “Excitation of Molecular Hydrogen in Seyferts: NGC 5506 and NGC 3081”, 2025, ApJ, 993, 217

‡ As contributing co-author:

- [20] Veenema, O., Rigopoulou, D., Thatte, N., et al., “Beyond the S-lines – New JWST H₂ O-line Excitation Diagnostics Point to X-ray Heating in the Active Galaxy NGC 3227”, 2026, MNRAS, submitted

- [19] Rosario, D., Haidar, H., Campbell, S., et al., “GATOS N: Extended circumnuclear dust emission in nearby Seyfert galaxies surveyed by JWST/MIRI”, 2026, MNRAS, in press
- [18] Donnan, F. R., García-Bernete, I., Rigopoulou, D., et al., “GATOS XIV: The first direct kinematic evidence of dusty outflows from AGN via PAH kinematics of local Seyfert galaxies with JWST”, 2026, MNRAS, 550, 1
- [17] Veenema, O., Thatte, N., Rigopoulou, D., et al., “Decoupling the AGN outflow and star-forming disk kinematics in the nuclear region of NGC 7582 with JWST NIRSpec and MIRI/MRS”, 2026, MNRAS, 548, 1
- [16] Hermosa Muñoz, L., González Fernández, J. R., Alonso-Herrero, A., et al., “Unveiling physical processes in local active galaxies: Unsupervised hierarchical clustering of JWST MIRI/MRS observations”, 2026, A&A, 708, 297
- [15] Haidar, H., Rosario, D., García-Bernete, I., et al., “GATOS XI. Excess dust heating in the Narrow Line Regions of nearby AGN revealed with JWST/MIRI”, 2026, MNRAS, 546, 1
- [14] Riffel, R. A., Colina, L., Costa-Souza, J. H., et al., “Impact of AGN and nuclear star formation on the ISM turbulence of galaxies: Insights from JWST/MIRI spectroscopy”, 2026, A&A, 705, 59
- [13] Veenema, O., Thatte, N., Rigopoulou, D., et al., “Shock-driven heating in the circumnuclear star-forming regions of NGC 7582: Insights from JWST NIRSpec and MIRI/MRS spectroscopy”, 2025, MNRAS, 544, 3361
- [12] Esposito, F., Alonso-Herrero, A., García-Burillo, S., et al., “GATOS X. Molecular gas clumpiness under the influence of AGN”, 2026, A&A, 705, 33
- [11] Campbell, S., Rosario, D., Haidar, H., et al., “GATOS IX. A Detailed Assessment and Treatment of Emission Line Contamination in JWST/MIRI Images of Nearby Seyfert Galaxies”, 2025, MNRAS, 544, 648
- [10] Lopez-Rodriguez, E., Ramos Almeida, C., Pereira-Santaella, M., et al., “GATOS VIII. On the physical origin of the extended MIR emission in AGN”, 2025, ApJ, 994, 206
- [9] González-Martín, O., Díaz-González, J. D., Martínez-Paredes, M., et al., “JWST reveals the diversity of nuclear obscuring dust in nearby AGN: nuclear isolation of MIRI/MRS data cubes and continuum spectral fitting”, 2025, MNRAS, 539, 2158
- [8] Esparza-Arredondo, D., Ramos Almeida, C., Audibert, A., et al., “Molecular gas stratification and disturbed kinematics in the Seyfert galaxy MCG-05-23-16 revealed by JWST and ALMA”, 2025, A&A, 693, 174
- [7] Fuller, L., Lopez-Rodriguez, E., García-Bernete, I., et al., “GATOS VII. The 20-214 μm imaging atlas of active galactic nuclei using SOFIA”, 2025, ApJS, 276, 64
- [6] García-Bernete, I., Rigopoulou, D., Donnan, F. R., et al., “GATOS V. Unveiling PAH survival and resilience in the circumnuclear regions of AGN with JWST”, 2024, A&A, 691, 162
- [5] Hermosa Muñoz, L., Alonso-Herrero, A., Pereira-Santaella, M., et al., “A biconical ionised gas outflow and evidence for positive feedback in NGC 7172 uncovered by MIRI/JWST”, 2024, A&A, 690, 350
- [4] García-Burillo, S., Hicks, K. S. E., Alonso-Herrero, A., et al., “Deciphering the imprint of AGN feedback in Seyfert galaxies: Nuclear-scale molecular gas deficits”, 2024, A&A, 689, 347
- [3] Haidar, H., Rosario, D., Alonso-Herrero, A., et al., “Dust beyond the torus: Revealing the mid-infrared heart of local Seyfert ESO 428-G14 with JWST/MIRI”, 2024, MNRAS, 532, 4645
- [2] Davies, R., Shimizu, T., Pereira-Santaella, M., et al., “GATOS: missing molecular gas in the outflow of NGC 5728 revealed by JWST”, 2024, A&A, 689, 263
- [1] Leist, M., Packham, C., Rosario, D., et al., “Deconvolution of JWST/MIRI Images: Applications to an Active Galactic Nucleus Model and GATOS Observations of NGC 5728”, 2024, AJ, 167, 96

★ **Other publications:**

- [3] Dressing, C. D., et al., “Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory”, 2026, arXiv:2608.11294
- [2] **Zhang, L.**, et al., “Deciphering The Launching of Multi-phase AGN-driven Outflows and Their (Spatially Resolved) Multi-scale Impact”, 2026, Astronomical Society of the Pacific Conference Series, 542, 161; arXiv:2508.01437
- [1] Skidmore, W., et al., “Thirty Meter Telescope International Observatory Detailed Science Case 2024”, 2024, arXiv:1505.01195v3